

# Understanding Score Propagation for Dynamic Objects

Two-time structure: internal time  $u$  vs. diffusion time  $t$

Internal research note

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## Abstract

We study score-based diffusion models when the data are *dynamic objects* (short trajectories) rather than static points. The conceptual difficulty is that two different “times” coexist: *internal time*  $u$  (frame index / physical evolution) and *diffusion time*  $t$  (Ornstein–Uhlenbeck corruption level). The correct two-time score object is

$$\mathbf{s}(\mathbf{x}, u, t) \equiv \nabla_{\mathbf{x}} \log p_t(\mathbf{x} \mid u),$$

and, for full trajectories, the path-space score

$$\mathbf{S}(\mathbf{x}_{0:T-1}, t) \equiv \nabla_{\mathbf{x}_{0:T-1}} \log p_t(\mathbf{x}_{0:T-1}).$$

The research goal is to formalize and test the hypothesis that the per-frame score components evolve across internal time via a simple propagator

$$\mathbf{S}_{:,u} \approx P(\mathbf{S}_{:,u-1}),$$

and to connect this  $P$  to the generator of the underlying physical dynamics, in an exactly solvable toy model (motion on a circle). The presentation follows and integrates [2, 1, 3].

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# 1 Motivation: From Static Data to Dynamic Objects

## 1.1 Trajectories as points in $\mathbb{R}^{dT}$ are formally correct but structurally wasteful

A trajectory is a sequence

$$\mathbf{a}_{0:T-1}^\mu = \{\mathbf{a}^\mu(u)\}_{u=0}^{T-1}, \quad \mathbf{a}^\mu(u) \in \mathbb{R}^d.$$

Concatenating all frames yields a vector in  $\mathbb{R}^{dT}$ , but this ignores redundancy: for many dynamic datasets, frames are highly correlated (e.g. rotation of a digit), so the intrinsic degrees of freedom scale *sublinearly* with  $T$ .

A useful quantitative proxy is the effective dimension of the concatenated covariance. Let  $C = \text{Cov}(\mathbf{a}_{0:T-1}) \in \mathbb{R}^{dT \times dT}$ . The participation ratio

$$d_{\text{eff}} = \frac{(\text{Tr} C)^2}{\text{Tr}(C^2)}$$

equals  $dT$  only if eigenvalues are flat; strong temporal correlations create a low-rank spectrum and  $d_{\text{eff}} \ll dT$ .

## 1.2 Two orthogonal axes: internal time $u$ vs. diffusion time $t$

We must keep separate:

- **Internal time  $u$ :** the (physical/semantic) evolution along the trajectory.
- **Diffusion time  $t$ :** the OU corruption level used to define the diffusion model.

Confusing these two is the main source of “fog” in the problem formulation [1, 3].

# 2 Mathematical Foundations (Careful and Explicit)

## 2.1 Langevin dynamics and Fokker–Planck

An Itô SDE in  $\mathbb{R}^d$  is

$$d\mathbf{x}_t = \mathbf{F}(\mathbf{x}_t, t) dt + \sqrt{2D} d\mathbf{W}_t, \tag{1}$$

with drift  $\mathbf{F}$  and diffusion coefficient  $D$ . The density  $p(\mathbf{x}, t)$  solves the Fokker–Planck equation

$$\partial_t p(\mathbf{x}, t) = -\text{div}(\mathbf{F}(\mathbf{x}, t) p(\mathbf{x}, t)) + D \Delta p(\mathbf{x}, t). \tag{2}$$

These are the standard tools used to derive forward/backward diffusion in score models [2].

## 2.2 Ornstein–Uhlenbeck forward diffusion (the modeling diffusion time)

We use the standard OU forward process in diffusion time  $t$ :

$$d\mathbf{x}_t = -\mathbf{x}_t dt + \sqrt{2} d\mathbf{W}_t. \tag{3}$$

It is exactly solvable:

$$\mathbf{x}_t = e^{-t} \mathbf{a} + \sqrt{1 - e^{-2t}} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d). \tag{4}$$

Define the two key scalars

$$\alpha_t = e^{-t}, \quad \Delta_t = 1 - e^{-2t}. \tag{5}$$

Then

$$p_t(\mathbf{x}) = \int_{\mathbb{R}^d} p_0(\mathbf{a}) \mathcal{N}(\mathbf{x}; \alpha_t \mathbf{a}, \Delta_t \mathbf{I}_d) d\mathbf{a},$$

a Gaussian smoothing of  $p_0$  [2].

### 2.3 Backward process and why the score appears

Let the score be

$$\mathbf{s}(\mathbf{x}, t) \equiv \nabla_{\mathbf{x}} \log p_t(\mathbf{x}).$$

Time reversal of the OU forward process yields a reverse-time dynamics whose drift contains the score term; in the notation used in [2],

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{x}(\tau) + 2\mathbf{s}(\mathbf{x}(\tau), t_f - \tau) + \boldsymbol{\eta}(\tau), \quad (6)$$

where  $\tau = t_f - t$  and  $\boldsymbol{\eta}$  is Gaussian white noise. The score is therefore *the force field* guiding denoising [2].

### 2.4 Exact score as Bayes-optimal denoiser (Tweedie identity)

Differentiating the mixture expression gives the exact identity (continuous-time OU)

$$\mathbf{s}(\mathbf{x}, t) = -\frac{\mathbf{x}}{\Delta_t} + \frac{\alpha_t}{\Delta_t} \mathbb{E}[\mathbf{a} \mid \mathbf{x}, t]. \quad (7)$$

Rearranging yields the posterior-mean denoiser

$$\mathbb{E}[\mathbf{a} \mid \mathbf{x}, t] = \frac{1}{\alpha_t} (\mathbf{x} + \Delta_t \mathbf{s}(\mathbf{x}, t)). \quad (8)$$

Thus: learning the score is equivalent to learning the Bayes-optimal denoiser [2].

## 3 Two-Time Structure of the Problem

### 3.1 Per-frame OU corruption at each internal index

Internal time  $u$  indexes frames. For each frame, apply OU corruption independently:

$$\mathbf{x}(u) = \alpha_t \mathbf{a}(u) + \sqrt{\Delta_t} \mathbf{z}(u), \quad \mathbf{z}(u) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \quad \text{i.i.d. across } u. \quad (9)$$

This defines, for each  $u$ , the marginal corrupted density

$$p_t(\mathbf{x} \mid u) = \int p_0(\mathbf{a} \mid u) \mathcal{N}(\mathbf{x}; \alpha_t \mathbf{a}, \Delta_t \mathbf{I}_d) d\mathbf{a}.$$

The corresponding **two-time score** is

$$\mathbf{s}(\mathbf{x}, u, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x} \mid u). \quad (10)$$

This is exactly the object emphasized in [1, 3].

### 3.2 Trajectory (path-space) score and why it differs from marginal scores

Let  $\mathbf{x}_{0:T-1} = (\mathbf{x}(0), \dots, \mathbf{x}(T-1)) \in \mathbb{R}^{dT}$ . The corrupted trajectory density is

$$p_t(\mathbf{x}_{0:T-1}) = \int p_0(\mathbf{a}_{0:T-1}) \prod_{u=0}^{T-1} \mathcal{N}(\mathbf{x}(u); \alpha_t \mathbf{a}(u), \Delta_t \mathbf{I}_d) d\mathbf{a}_{0:T-1}. \quad (11)$$

The path score is

$$\mathbf{S}(\mathbf{x}_{0:T-1}, t) = \nabla_{\mathbf{x}_{0:T-1}} \log p_t(\mathbf{x}_{0:T-1}),$$

with components  $\mathbf{S}_{:,u} = \nabla_{\mathbf{x}(u)} \log p_t(\mathbf{x}_{0:T-1})$ . Because  $p_0(\mathbf{a}_{0:T-1})$  couples frames,  $\mathbf{S}_{:,u}$  depends on *neighboring frames* through the posterior, which is the mechanism behind “score propagation across  $u$ ” [1, 3].

## 4 Exactly Solvable Toy Model: Motion on a Circle

We use the circle model defined in [1, 3]:

$$\mathbf{a}(u) = (x(u), y(u)) = (r(u) \cos \theta(u), r(u) \sin \theta(u)).$$

The internal-time Langevin SDEs are

$$dr(u) = (1 - r(u)) du + \sqrt{2T_r} dB_r(u), \quad (12)$$

$$d\theta(u) = \omega du + \sqrt{2T_\theta} dB_\theta(u). \quad (13)$$

Radial confinement corresponds to the potential

$$V(r) = \frac{1}{2}(r - 1)^2,$$

so  $1 - r = -\partial_r V$  is a restoring force toward  $r = 1$ . Angular motion is approximately deterministic when  $T_\theta \ll \omega$ . This model gives curvature, a known generator, and analyzable score geometry [1, 3].

## 5 Per-Frame Diffusion and the Exact Score

At fixed  $u$ , the corrupted density is a Gaussian smoothing of  $p_0(\cdot | u)$ . The exact marginal score satisfies (conditional version of (7))

$$\mathbf{s}(\mathbf{x}, u, t) = -\frac{\mathbf{x}}{\Delta_t} + \frac{\alpha_t}{\Delta_t} \mathbb{E}[\mathbf{a}(u) | \mathbf{x}, u, t]. \quad (14)$$

**Limits in diffusion time.** As  $t \downarrow 0$ ,  $\alpha_t \rightarrow 1$  and  $\Delta_t \rightarrow 0$ , hence  $p_t(\cdot | u) \rightarrow p_0(\cdot | u)$  and the score approaches the clean score. As  $t \rightarrow \infty$ ,  $p_t(\cdot | u) \rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I})$  and the score approaches  $-\mathbf{x}$ .

## 6 Central Hypothesis: Temporal Evolution of the Score

Following [1, 3], define the per-frame score component of the *trajectory* score:

$$\mathbf{S}_{;u}(\mathbf{x}_{0:T-1}, t) \in \mathbb{R}^d.$$

The hypothesis is that there exists a (possibly approximately) stationary propagator  $P$  such that

$$\mathbf{S}_{;u}(\mathbf{x}_{0:T-1}, t) \approx P\left(\mathbf{S}_{;u-1}(\mathbf{x}_{0:T-1}, t)\right). \quad (15)$$

In rotation-dominated regimes (circle model),  $P$  is expected to be close to a rotation operator acting on the score field, i.e.  $P \approx R_{\Delta\theta}$  in 2D [1, 3].

### 6.1 Connection to transfer/Koopman viewpoint (why $P$ should exist)

Internal-time dynamics transports densities; scores (gradients of log-densities) transport accordingly. In the simplest case where  $\mathbf{a}(u) = g(\mathbf{a}(u - 1))$  is an invertible map, the score field obeys the exact transformation law

$$\nabla \log p_u(\mathbf{x}) = J_{g^{-1}}(\mathbf{x})^\top \nabla \log p_{u-1}(g^{-1}(\mathbf{x})) + \nabla \log |\det J_{g^{-1}}(\mathbf{x})|.$$

For rigid rotations,  $J_{g^{-1}}$  is constant orthogonal and the Jacobian-determinant term vanishes, giving pure rotation of the score field.

## 7 Relation to Board Exercises

We restate the three board exercises (integrated form) as in [3].

### 7.1 Exercise 1: 2D discrete rotation and continuous limit

Discrete internal-time dynamics:

$$\mathbf{a}(u+1) = R_{\Delta\theta}\mathbf{a}(u) + \boldsymbol{\xi}_u, \quad \boldsymbol{\xi}_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_2).$$

This makes the internal-time “ground truth” close to a rotation, hence  $P$  should be close to a rotation in the clean regime.

### 7.2 Exercise 2: 1D generic one-step kernel and joint score components

For  $x_{u+1} = x_u + c + \eta$  with  $\eta \sim \mathcal{N}(0, \sigma^2)$ , the joint density

$$p(x_u, x_{u+1}) = p_0(x_u) \mathcal{N}(x_{u+1}; x_u + c, \sigma^2)$$

gives joint score components

$$S_0 = \partial_{x_u} \log p = \partial_{x_u} \log p_0(x_u) + \frac{x_{u+1} - (x_u + c)}{\sigma^2}, \quad S_1 = \partial_{x_{u+1}} \log p = -\frac{x_{u+1} - (x_u + c)}{\sigma^2}.$$

This makes explicit that temporal coupling appears in the trajectory score as “spring-like” forces tying adjacent frames.

### 7.3 Exercise 3: fixed internal index, varying diffusion time

Fix  $u$  and increase  $t$ ; the OU smoothing scale increases, and the score magnitude decreases accordingly, consistent with the OU limits.

## 8 Why This Is Not Standard Video Diffusion

Standard video diffusion often learns a large joint network over space–time. Here, instead, the aim is *operator identification*: extract a simple  $P$  (and ideally its generator) that reflects internal-time physics, and test it quantitatively (spectrum, stability across  $t$ , extrapolation), as emphasized in [1].

## 9 Research Roadmap (Fog $\rightarrow$ Sun)

- **Analytic ground truth:** use exact OU identities from [2] (score as posterior mean/denoiser), and the circle model from [1, 3].
- **Simulation:** generate clean trajectories by (12)–(13), then corrupt per-frame via (9).
- **Score estimation without NNs:** compute empirical mixture scores at fixed  $u$  from samples (exact formula in [2]) to visualize fields.
- **Learn/fit  $P$ :** start with linear  $P$  (e.g. least squares on score components) and test whether eigenvalues encode rotation (complex phases) and whether  $P$  is stable across  $t$ .

References: [2, 1, 3].

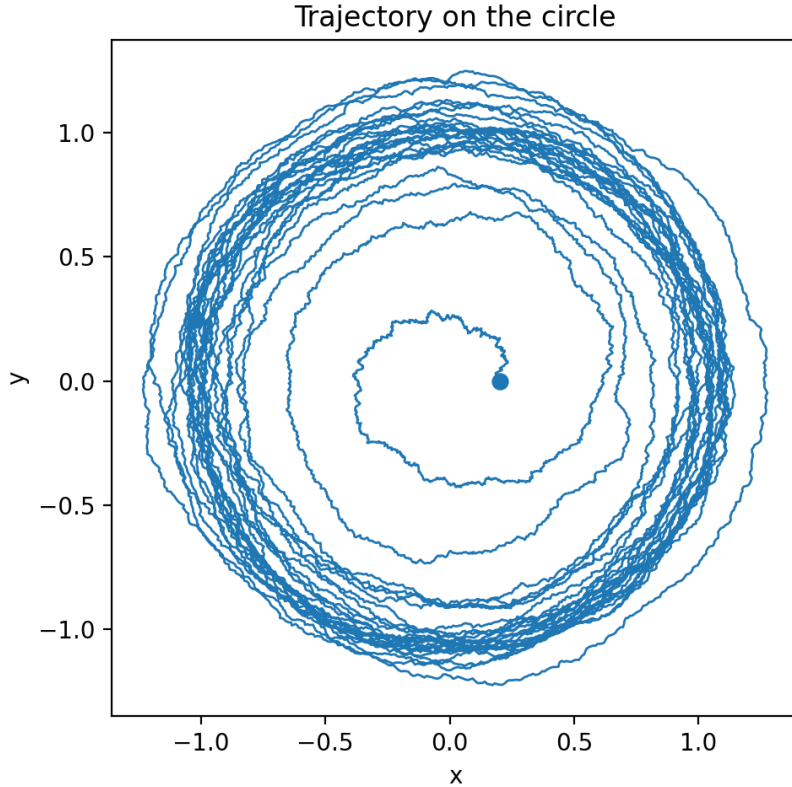


Figure 1: A single internal-time trajectory on the noisy ring.

## 10 Visualization Requirements: Figures and Python Code

All images are saved under the folder `figures/`. The code uses only `numpy` and `matplotlib`.

### 10.1 Figure 1: Trajectory on the circle

### 10.2 Figure 2: Radial relaxation over internal time

### 10.3 Figure 3: Score vector field at fixed $(u, t)$

We compute an empirical marginal score field from samples at a fixed internal index  $u$  using the exact mixture-score formula (empirical score) [2].

### 10.4 Figure 4: Rotation of the score field across internal time

### 10.5 Figure 5: Effect of increasing diffusion time on score magnitude

## References (project sources)

## References

- [1] Giovanni & Marco. *Score-Based Generative Diffusion Models for Dynamic Objects: Thesis Direction Notes*. January 2026.
- [2] Marc Mézard. *Generative diffusion updated notes*. December 2025.
- [3] Board notes. *Board exercises: score evolution across internal time  $u$  and diffusion time  $t$* . 2026.

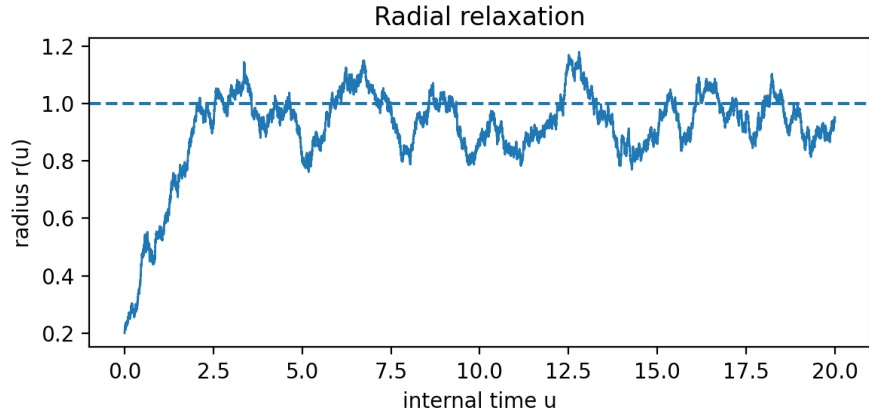


Figure 2: Radial coordinate  $r(u)$  relaxes toward 1 with fluctuations controlled by  $T_r$ .

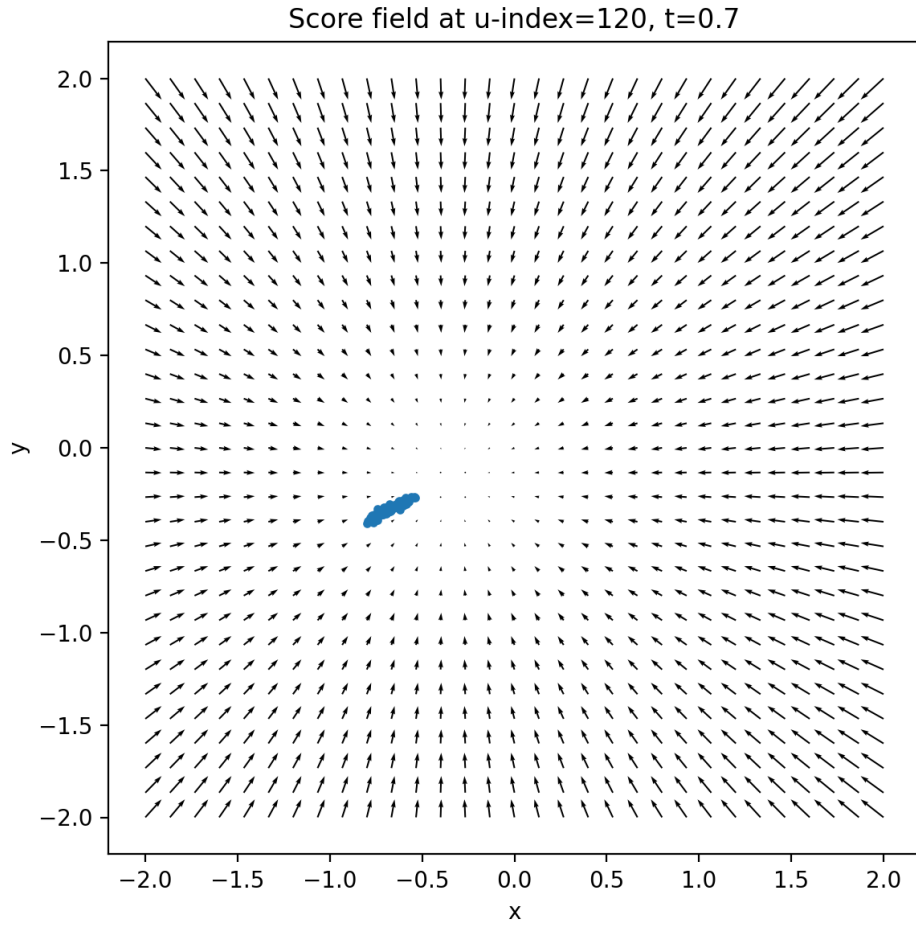


Figure 3: Empirical score field at a fixed internal index and diffusion time.



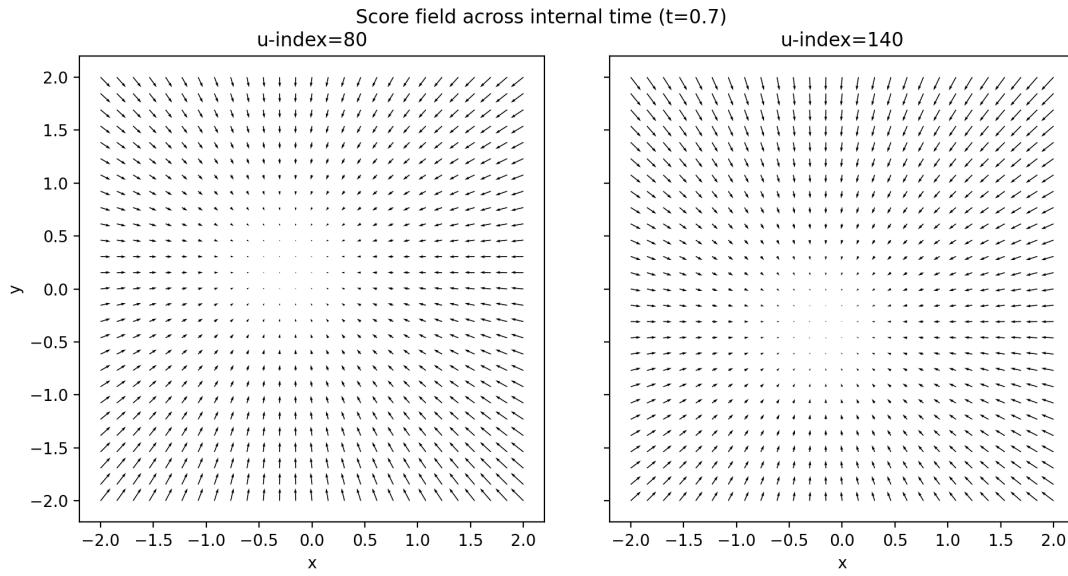


Figure 4: Score fields at two internal indices; in rotation-dominated regimes the field should approximately rotate across  $u$ .

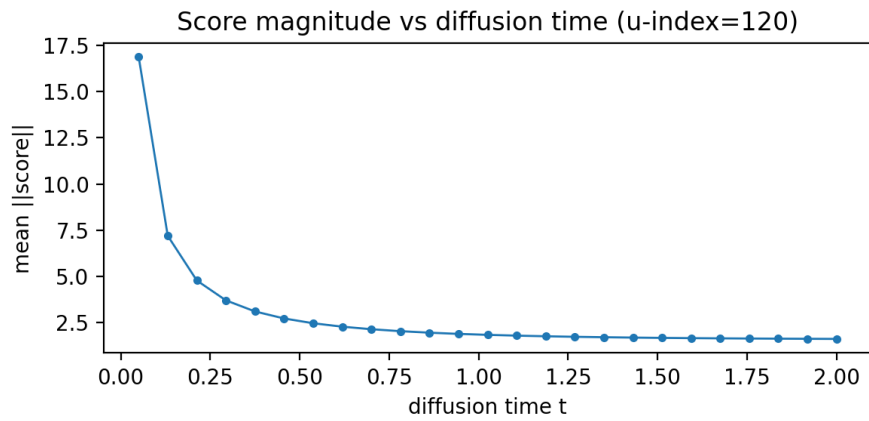


Figure 5: Mean score magnitude on a grid versus diffusion time.